

Predictors of Multiple Seizures in a Cohort of Children Prospectively Followed from the Time of Their First Unprovoked Seizure

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The objective of this study was to assess the risk of multiple recurrences after an initial seizure recurrence in childhood. In a prospective study, 407 children were followed for a mean of 9.6 years from the time of their first unprovoked seizure. Data regarding each seizure recurrence were obtained and analyzed using statistical methods for survival analysis. The cumulative risk of a second seizure was 29%, 37%, 43%, and 46% at 1, 2, 5, and 10 years, respectively. Of the 182 children who experienced a second seizure, 131 (72%) experienced a third seizure, 105 (58%) have had 4 or more seizures, and 52 (29%) have experienced 10 or more seizures. The cumulative risk of a third seizure was 57%, 63%, and 71% at 1, 2, and 5 years, respectively, after the second seizure. After a third seizure, the cumulative risk of another seizure was 69%, 72%, and 81% at 1, 2, and 5 years, respectively. After a second seizure, factors associated with an increased risk of additional recurrences included a remote symptomatic etiology (rate ratio = 1.7) and the occurrence of a second seizure within 6 months of the first seizure (rate ratio = 1.7). After a second seizure, the risk of subsequent seizures was greater than 50% even in the lowest risk group. With the exception of etiology, factors associated with an increased risk of multiple recurrences after the initial seizure were different than those associated with multiple recurrences after a second seizure. Factors associated with multiple recurrent seizures may be different than those associated with an initial recurrence. As most patients who experience a second seizure experience further seizures, these data suggest that two seizures are a sufficient epidemiological criterion for the definition of epilepsy.

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Most children with a single unprovoked seizure do not experience a recurrence.^{1–16} Only one study in a primarily adult population has examined the risk of further seizures after a second seizure.¹⁷ The data from the adult study suggest that a single seizure may be an isolated event, but once two seizures have occurred, the likelihood of further seizures is 70% or greater, and one is thus dealing with a chronic process.^{17,18} We have followed 407 children from the time of their first unprovoked seizure. Details of the cohort and the risk of seizure recurrence after the initial seizure have been published previously.^{1,2} This analysis focuses on the risk of subsequent recurrences after an initial seizure recurrence and risk factors for further recurrences in this cohort.

Materials and Methods

Subjects

In a prospective cohort study, 407 children with a first unprovoked seizure who were seen at Montefiore Medical Center, Jacobi Medical Center, North Central Bronx Hospital, or the private practices of the authors between October 1983 and August 1992 were enrolled and followed until September 1, 1998. Eligible candidates for the study were patients aged 1 month to 19 years who presented with their first unprovoked afebrile seizure. Details of the inclusion and exclusion criteria for this cohort as well of the initial evaluation have been reported previously.^{1,2}

At the time of the initial visit, informed consent was obtained from the parent and child. Details, including seizure characteristics, duration, number of seizures in 24 hours, and any treatment given, were collected. Additional information

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regarding prior provoked seizures, prior neurological insults, birth history, and family history was also coded. A physical and neurological examination was performed on all children. Electroencephalograms (EEGs) were scheduled for all patients. Neuroimaging studies were performed when clinically indicated. Etiology and seizure type were classified in accordance with International League Against Epilepsy criteria.^{18–20}

Follow-Up

After enrollment, subjects were followed by telephone interviews at least every 3 months for ascertainment of any seizure recurrence. In those children with a recurrence, records of any emergency medical care were reviewed and the children were re-evaluated. A recurrence was defined as any unprovoked seizure occurring more than 24 hours after the first seizure. The mean follow-up period was 9.6 years. Of the 407 subjects, 391 (96%) have been followed for more than 2 years and 372 (91%) for more than 5 years. The 16 subjects followed less than 2 years include 1 patient who died and 15 lost to follow-up.

Analysis

As the length of the follow-up interval influences the probability of observing a recurrence, the statistical methods took account of the variable length of follow-up for each child.^{21–27} Risks of a first, second, and ninth recurrence at 6 months and at 1, 2, 5, and 10 years after the initial seizure as well as risks of a second, third, and ninth recurrence at 6 months and at 1, 2, 5, and 10 years after a first recurrence (second seizure) were determined by the product-limit method, and the results were displayed as Kaplan-Meier curves. Ninety-five percent confidence intervals (CIs) for the Kaplan-Meier estimates of recurrence risks were calculated at 1, 2, 5, and 10 years using an approximate Greenwood formula for the SE.^{24,27}

The Cox proportional hazards model was used to obtain crude and adjusted rate ratios for each independent variable and for the multivariate analysis.^{21,24–27} To determine the appropriateness of the proportional hazards assumption (ie, constant rate ratio over time), a term was included for the independent variable multiplied by the log of time (in years) in each unadjusted analysis. Where there was evidence for lack of proportionality over time based on either visual inspection of the log(–log) survival plots or a test of time dependence using the log of time, we examined cutoff points at 3-month intervals during the first year and at 1 and 2 years as well as at later points if necessary to determine whether there was an abrupt change in the rate ratio at one point in time or whether the rate ratio changed continuously over time. A single cutoff point was used unless the continuous log representation provided a significantly better ($p < 0.05$) fit for the statistical model.²⁷

Results

Overall Recurrence Risk

Of the 407 subjects in the study, 182 (45%) have experienced a recurrence. The overall product-limit estimate of recurrence was 22% at 6 months (95% CI: 18%, 26%), 29% at 1 year (95% CI: 25%, 33%),

37% at 2 years (95% CI: 33%, 42%), 43% at 5 years (95% CI: 38%, 48%), and 46% at 10 years (95% CI: 41%, 51%) (Fig 1). The median time to recurrence was 6.2 months (range, 0.03 months to 10.1 years).

Of those who experienced a second seizure, 131 (72%) experienced a third seizure, 105 (58%) experienced 4 or more seizures, and 52 (29%) experienced 10 or more seizures (Table 1, Fig 2). The primary analysis is the risk of another seizure after a second seizure has occurred. We also analyzed the risk of three or more seizures using only the data available at the time of first seizure.

Risk of a Third Seizure after a Second Seizure

The 182 children who had a second seizure have been followed for a mean of 8.4 years after their second seizure (see Table 1). The risk of a third seizure was 57%, 63%, and 72% at 1, 2, and 5 years, respectively, after the second seizure (see Fig 2). After a third seizure, the cumulative risk of another seizure was 66%, 70%, and 81% at 1, 2, and 5 years, respectively.

UNIVARIATE ANALYSIS. Univariate analysis of factors associated with a differential risk of subsequent recurrences after a second seizure (first recurrence) is shown in Table 2. Remote symptomatic etiology (Fig 3) and an interval of less than 6 months between the first and second seizures were associated with an increased risk of having a third seizure. The increased recurrence risk associated with a remote symptomatic etiology is not time-dependent and persists throughout the analysis period. In contrast, an interval less than 6 months between the first and second seizures was associated with an increased recurrence risk only during the first 3 months after the second seizure (relative risk [RR] = 2.99; 95% CI: 1.75, 5.12; $p = 0.0001$), after which it

Fig 1. Probability of seizure recurrences after a first unprovoked seizure ($N = 407$). Kaplan-Meier curves for cumulative risk of a second (R1), third (R2), fourth (R3), and tenth (R9) seizure in children who have had a first unprovoked seizure calculated from the time of the first seizure.

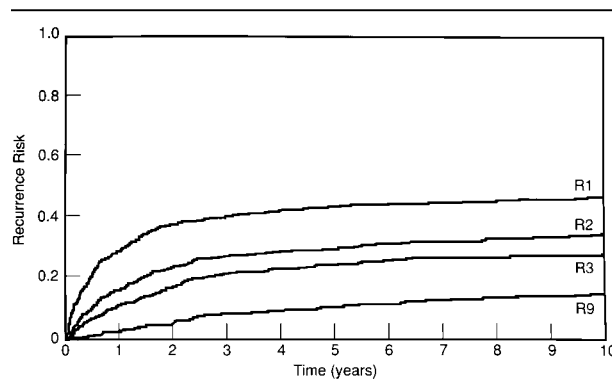


Table 1. Characteristics of Cohort of 407 Children Followed from Time of First Unprovoked Seizure

	Overall	Cryptogenic/Idiopathic Etiology	Remote Symptomatic Etiology
No. of children	407	342	65
Mean age at first seizure (yr)	6.8	6.8	7.0
Mean follow-up after first seizure (yr)	9.6	9.6	9.7
Recurrent seizures	182 (45%)	135 (39%)	47 (72%)
Mean follow-up after second seizure (yr)	8.4	8.4	8.6
Total number of seizures to date			
1 (no recurrences)	225 (55%)	207 (60%)	18 (28%)
2 (one recurrence)	51 (13%)	46 (13%)	5 (8%)
3	26 (6%)	18 (5%)	8 (12%)
4	13 (3%)	10 (3%)	3 (5%)
5	12 (3%)	10 (3%)	2 (3%)
6–9	28 (7%)	20 (6%)	8 (12%)
≥10	52 (13%)	31 (10%)	21 (32%)

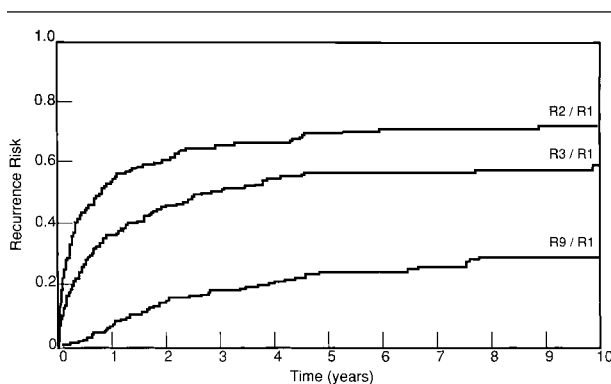


Fig 2. Probability of subsequent seizure recurrences after a second seizure (first recurrence) ($N = 182$). Kaplan-Meier curves for cumulative risk of a third (R2), fourth (R3), and tenth (R9) seizure in children who have had two unprovoked seizures calculated from the time of the second seizure.

no longer influenced recurrence risk ($RR = 0.92$; 95% CI: 0.53, 1.57; $p = 0.002$ for difference in the RRs).

Eighty (44%) of the 182 children were treated with anti-epileptic drugs (AEDs) after the first recurrence. On univariate analysis, treatment after the second seizure (first recurrence) did not affect the risk of subsequent seizures. However, children with remote symptomatic seizures were more likely to be treated after a first recurrence (60%) than children with cryptogenic first seizures (38%) ($p = 0.01$), as were children whose first recurrence occurred within 6 months (56%) compared with children whose first recurrence occurred after 6 months (33%) ($p < 0.01$).

An abnormal EEG ($RR = 1.08$; 95% CI: 0.81, 1.67; $p = 0.41$) and a seizure occurring while asleep ($RR = 0.86$; 95% CI: 0.61, 1.23; $p = 0.41$), which were associated with an increased recurrence risk after a first seizure, were not associated with an increased risk of further seizures after a second seizure. The occurrence of status epilepticus ($RR = 0.84$; 95% CI: 0.51,

1.41; $p = 0.51$) or of multiple seizures as the presenting first seizure episode was also not associated with an increased risk of subsequent recurrences.

MULTIVARIATE ANALYSIS. On multivariate analysis (Table 3), remote symptomatic etiology and an interval less than 6 months between the first and second seizures remain significant. After adjusting for etiology and time to second seizure, treatment after the second seizure is also highly significant and is associated with a greater than 50% reduction in the risk of subsequent seizures for the first 3 months. No other variables entered the model.

Predictors of Multiple Recurrences after a Second Seizure

The univariate and multivariate analyses for the occurrence of a fourth and tenth seizure after a second seizure has occurred are shown in Tables 2 and 3. Etiology and an interval less than 6 months between the first and second seizures have a similar association with the risk of a fourth seizure as they did with the risk of a third seizure. Treatment after the second seizure is also important after adjusting for etiology, and the effect does not vary significantly over time (see Table 3). In addition, a history of prior febrile seizures and having multiple seizures in 1 day as the initial seizure have a modest but statistically significant association with an increased risk of a fourth seizure after a second seizure (see Tables 2 and 3).

In predicting the occurrence of 10 or more seizures after a second seizure, a remote symptomatic etiology and an interval less than 6 months between the first and second seizures are significantly associated with an increased risk of having 10 or more seizures in both univariate and multivariate analyses (see Tables 2 and 3). The effect of the interval of less than 6 months is largely limited to the first year after the second seizure.

Table 2. Risk of Multiple Recurrences following a Second Seizure: Proportional Hazards Model (N = 182)^a

Risk Factor	Risk of Third Seizure (n = 131)			Risk of Fourth Seizure (n = 105)			Risk of Ten Seizures (n = 52)		
	Rate Ratio	95% CI	p	Rate Ratio	95% CI	p	Rate Ratio	95% CI	p
Remote symptomatic etiology	1.69	(1.17, 2.45)	0.005	1.65	(1.10, 2.49)	0.02	2.20	(1.26, 3.83)	0.005
Time to second seizure < 6 mo ^a	1.60 ^a	(1.14, 2.26)	0.007	2.10 ^a	(1.42, 3.10)	0.0002	2.10 ^a	(1.19, 3.72)	0.01
Treatment after second seizure ^a	0.92 ^a	(0.65, 1.29)	0.62	0.86	(0.59, 1.27)	0.45	1.19	(0.69, 2.05)	0.53
Abnormal electroencephalogram	1.08	(0.81, 1.67)	0.41	1.13	(0.77, 1.68)	0.53	0.90	(0.52, 1.55)	0.70
Prior febrile seizures	1.32	(0.88, 1.98)	0.17	1.60	(1.04, 2.47)	0.04	1.42	(0.77, 2.62)	0.26
Todd's paresis	0.94	(0.51, 1.74)	0.84	1.30	(0.69, 2.42)	0.42	2.05	(0.97, 4.36)	0.06
>1 seizure within 24 hr of the first seizure	1.23	(0.82, 1.83)	0.31	1.57	(1.02, 2.40)	0.04	1.06	(0.56, 2.03)	0.85

^aAnalysis is shown without effect of time-dependent covariates. Variables for which time-dependent covariates are significant are indicated, and the effects of the time-dependent covariates on these variables are shown in the multivariable analysis in Table 3.

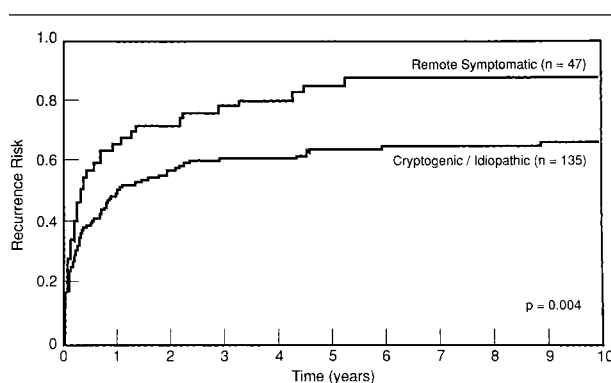


Fig 3. Probability of subsequent seizure recurrences after a second seizure in children with cryptogenic/idiopathic and remote symptomatic seizures (N = 182). Kaplan-Meier curves.

Predictors of Multiple Recurrences at the Time of Initial Seizure

A separate analysis of predictors of multiple recurrences based on information available at the time of the first seizure was performed (Tables 4 and 5). The Kaplan-Meier curves illustrating the risk for multiple seizures after the initial seizure are shown in Figure 1. As previously reported,^{1,2} remote symptomatic etiology, abnormal EEG, being asleep at the time of the first seizure, a history of prior febrile seizures, and the presence of a Todd's paresis were all associated with an increased risk of a second seizure in both univariate and multivariate analyses. An analysis of the time dependence of the effect showed that the influence of etiology on recurrence risk, although present throughout, was significantly stronger after the first 2 years (see Table 5). There was no evidence that the strength of the association with recurrence risk was time-dependent for any of the other variables.

Not surprisingly, factors associated with a differential risk of a second seizure are also associated with an increased risk of a third seizure. Etiology and the EEG as

well as a history of prior febrile seizures were significant and independent predictors of the risk of a third seizure at the time of the first seizure (see Tables 4 and 5). In predicting the risk of having 10 or more seizures at the time of the first seizure, a remote symptomatic etiology and the presence of a Todd's paresis were the only two significant predictors (see Tables 4 and 5). There was no evidence of departure from proportionality over time for any of the variables, including etiology, in the analysis of predictors of 10 or more seizures at the time of the first seizure.

Discussion

How Many Seizures Is Epilepsy?

After two seizures, the risk of subsequent seizures goes up to 70% or more. Although factors such as etiology and the interval between the first and second seizures are associated with a differential risk of recurrence, the recurrence risk is greater than 60% in all subgroups. Furthermore, the risk of an additional seizure does not substantially change after a third seizure. This supports the epidemiological definition of epilepsy as two or more unprovoked seizures.¹⁸ The overall recurrence risk after a second seizure in this study is similar to that recently reported in a largely adult population.¹⁷

Although from an epidemiological standpoint, the data justify requiring at least two seizures for the definition of epilepsy, this may not always be the case in other settings. The risk of a seizure recurrence in a child with a first seizure of remote symptomatic etiology or cryptogenic/idiopathic etiology and an epileptiform EEG is quite comparable to the recurrence risk after a second seizure.¹⁻³ Therefore, if one is designing a drug trial for new-onset seizures, it may be quite rational to include children with one seizure and additional risk factors for recurrence.²⁸ In particular, the recurrence risk in this cohort in children with benign Rolandic epilepsy, who all have a characteristic epilep-

Table 3. Risk Factors for Subsequent Seizures after a Second Seizure: Multivariable Analysis Using Cox Proportional Hazards Model^a

Risk Factor	Risk of Third Seizure (n = 131)			Risk of Fourth Seizure (n = 105)			Risk of Ten Seizures (n = 52)		
	Rate Ratio	95% CI	p	Rate Ratio	95% CI	p	Rate Ratio	95% CI	p
Remote symptomatic etiology	1.69	(1.16, 2.47)	0.007	1.74	(1.14, 2.66)	0.01	2.13	(1.22, 3.71)	0.008
Second seizure within 6+ mo									
In first 3 months (12 mo for 10 seizures)	4.00	(2.28, 7.00)	<0.0001	5.07	(2.19, 11.7)	0.0002	6.94	(1.56, 30.7)	0.01
After 3 months (12 mo for 10 seizures)	0.86	(0.49, 1.50)	<0.0001 ^b	1.92	(0.83, 4.46)	0.05 ^b	1.43	(0.32, 6.33)	0.06 ^b
Treatment after second seizure ^a	—	—	—	0.56	(0.37, 0.85)	0.006	—	—	—
First 3 mo	0.37	(0.22, 0.63)	0.0003	—	—	—	—	—	—
After 3 mo	1.28	(0.75, 2.18)	0.001 ^b	—	—	—	—	—	—
>1 seizure within 24 hr of the first seizure	—	—	—	1.73	(1.12, 2.68)	0.01	—	—	—
Prior febrile seizures	—	—	—	1.56	(1.00, 2.44)	0.05	—	—	—

^aEffect of variable is best modeled with a time-dependent covariate.

^bProbability value on interaction represented by time-dependent covariate. This indicates that the RR after 3 (or 12) months have passed after the initial recurrence is significantly different from the RR before that time.

Table 4. Risk of Seizure Recurrences after a First Unprovoked Seizure: Proportional Hazards Model (N = 407)

Risk Factor	Risk of Second Seizure (n = 182)			Risk of Third Seizure (n = 131)			Risk of Ten Seizures (n = 52)		
	Rate Ratio	95% CI	p	Rate Ratio	95% CI	p	Rate Ratio	95% CI	p
Remote symptomatic etiology ^a	2.32 ^a	(1.67, 3.24)	<0.0001	3.20	(2.22, 4.63)	<0.001	4.04	(2.32, 7.03)	<0.0001
Abnormal electroencephalogram	2.16	(1.60, 2.91)	<0.0001	2.15	(1.51, 3.05)	<0.0001	1.72	(1.00, 2.98)	0.05
Seizure while asleep	1.61	(1.20, 2.17)	0.0002	1.37	(0.97, 1.95)	0.07	1.28	(0.73, 2.24)	0.38
Prior febrile seizures	1.54	(1.08, 2.20)	0.02	1.79	(1.19, 2.67)	0.005	1.92	(1.04, 3.55)	0.04
Todd's paresis	1.81	(1.08, 3.03)	0.02	1.49	(0.81, 2.77)	0.20	2.95	(1.39, 6.27)	0.005
>1 seizure within 24 hr of the first seizure	1.04	(0.73, 1.47)	0.83	1.15	(0.76, 1.72)	0.48	1.05	(0.55, 2.01)	0.87

^aAnalysis is shown without effect of time-dependent covariates. The effect of remote symptomatic etiology is time-dependent. The effects of the time-dependent covariate on this variable are shown in the multivariable analysis in Table 5.

Table 5. Risk Factors for Seizure Recurrences after a First Unprovoked Seizure: Multivariable Analysis Using Cox Proportional Hazards Model

Risk Factor	Risk of Second Seizure (n = 182)			Risk of Third Seizure (n = 131)			Risk of Ten Seizures (n = 52)		
	Rate Ratio	95% CI	p	Rate Ratio	95% CI	p	Rate Ratio	95% CI	p
Remote symptomatic etiology ^a				2.64	(1.82, 3.84)	<0.0001	4.10	(2.35, 7.15)	<0.0001
First 24 mo	1.61 ^a	(1.11, 2.34)	0.01						
After 24 mo	4.00 ^a	(2.75, 5.83)	0.03 ^b						
Abnormal electroencephalogram	1.95	(1.44, 2.64)	<0.0001	1.93	(1.35, 3.84)	0.0003			
Prior febrile seizures	1.63	(1.13, 2.33)	0.008	1.75	(1.17, 2.63)	0.007			
Seizure while asleep	1.41	(1.04, 1.90)	0.03						
Todd's paresis	1.68	(1.00, 2.82)	0.05				3.07	(1.44, 6.53)	0.004

^aEffect of variable is best modeled with a time-dependent covariate.

^bProbability value on interaction represented by time-dependent covariate. This indicates that the RR after 3 (or 12) months have passed after the initial recurrence is significantly different from the RR before that time.

tiform EEG,²⁹ is quite similar after one seizure to the recurrence risk after two seizures. Including such patients may make it easier to conduct placebo-controlled trials.³⁰

Predicting 10 or More Seizures

Another area of interest is early identification of patients who will go on to have many seizures. These children would be candidates not only for early treatment with AEDs but possibly for more aggressive early intervention.^{31–33} Few children who present with a first unprovoked seizure develop intractable epilepsy. This is partly because the syndromes most likely to progress to intractable epilepsy rarely present with a single unprovoked seizure.³⁴ If one examines the subgroup with 10 or more seizures in this cohort, which constitutes only 13% of the cohort, only remote symptomatic etiology and early initial recurrence are associated with an increased risk of 10 or more seizures after a second seizure. Remote symptomatic etiology is consistently associated with a higher recurrence risk after a first seizure,^{1–5,6,10–13} with a higher risk of recurrence after AED withdrawal in patients who are seizure-free,^{35,36} with a lower probability of attaining remission,^{37–39} and with a higher risk of developing intractable epilepsy.³¹ It is also a risk factor in this population for having many seizures, as 32% of the 65 children with a remote symptomatic first seizure in this cohort experienced 10 or more seizures compared with 9% of children with a cryptogenic/idiopathic first seizure ($p < 0.001$).

After adjustment for etiology, treatment reduces recurrence risk by half but only for the first 3 months. The reduction of recurrence risk in this observational study is comparable to the reduction reported in randomized placebo-controlled treatment trials after a first unprovoked seizure in children and adults.^{14,15} Those studies did not analyze whether the effect is time-dependent. A time-dependent effect for the efficacy of AED treatment was also observed in the Medical Research Council study of discontinuing AEDs in patients with epilepsy who were seizure-free for 2 or more years.⁴⁰ In that study, which randomized subjects to either discontinue medications or remain on AEDs, the recurrence risk was twice as high in those who discontinued AEDs, but the increased risk only persisted for a finite period of time, after which the 2 groups had similar recurrence risks.⁴¹ In our observational study, treatment did not influence the risk of having 10 or more seizures. As the probability of a clinical decision to initiate AED therapy substantially increased with each recurrence, this finding is somewhat difficult to interpret. These findings are similar to those of the randomized Italian study of treatment after a first seizure,⁴¹ which found no effect of delaying treatment on long-term outcome in the sense that delaying treatment until at

least the third seizure did not affect long-term outcome as measured by the risk of having 10 or more seizures.

The association between the time interval between the first and second seizures is not surprising, as it may reflect how aggressive the underlying process is. The effect is time-dependent and transient. Initial seizure frequency, which is inversely related to the interval between seizures, has been associated with the probability of attaining remission in patients with childhood-onset epilepsy.^{33,39}

Factors such as a Todd's paresis, prior febrile seizures, and the occurrence of multiple seizures within 24 hours are of marginal significance in the overall analysis, although they may contribute to some of the multiple analyses performed. In contrast to a recent abstract suggesting that the occurrence of multiple seizures in 1 day is associated with a worse prognosis,⁴² these data support the International League Against Epilepsy position¹⁸ and other reports^{1,2,5,43,44} that multiple seizures in a 24-hour interval have the same prognostic significance as a single isolated seizure.

When Is Treatment with AEDs Indicated?

In general, the decision to initiate AED therapy is based on weighing the relative risks of further seizures versus the risks of AED therapy.⁴⁵ Given the high risk of recurrent seizures, most adult neurologists initiate treatment after a second seizure.¹⁷ In children, in addition to the recurrence risks, treatment decisions should take account of whether the seizures are part of a benign self-limited syndrome such as benign Rolandic epilepsy as well as seizure frequency and duration.^{45–47} The authors often do not treat otherwise normal children with infrequent brief seizures even after two or three seizures have occurred.^{45,46} Data from randomized clinical trials of children and adults who present with a first unprovoked seizure have shown that AED therapy reduces the risk of seizure recurrence by approximately half.^{14,15} Early treatment does not affect the probability of attaining long-term remission, however.⁴¹ Our data further suggest that the effect on recurrence may be of relatively limited duration. Studies in developing countries, where treatment delays were due to the unavailability of AEDs, have shown no difference in response rate in those with many prior seizures compared with new-onset patients.^{48,49} In general, the prognosis seems to be a function of the specific epilepsy syndrome. AED therapy suppresses seizures but does not alter the underlying course of the illness.^{47,50–52} The decision to treat should therefore be made on the grounds that the patient has had a sufficient number of events to justify therapy and not with the hope of somehow preventing the development of "chronic" epilepsy.⁵³ Although the probability of a seizure recurrence seems to be the same in children and adults, the relative risks and benefits of AED therapy,

which take account of not only the statistical probability of further seizures but also the possible consequences of another seizure and the potential adverse effects of AED therapy, are quite different in children than in adults.^{45,46,54}

Conclusions

Most children who experience two seizures experience further seizures. Thus, the definition of epilepsy as two or more unprovoked seizures occurring more than 24 hours apart is appropriate for epidemiological studies. Etiology is the most important factor influencing long-term prognosis in this group of children. Treatment decisions in these children need to be individualized based on the statistical recurrence risk and the relative risks of both seizures and AED therapy.

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